STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: MAC-03 | TAG: 003

MAC-03: Bridgeport Series I Manual Knee Mill #3

Facility Zone: Zone 1: Manual Machining (MAC) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

CRITICAL ENTANGLEMENT WARNING: STRICTLY NO GLOVES WHILE SPINDLE / BLADE IS ROTATING

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses
2. Closed-Toe Shoes
3. Hair & Loose Clothing Secured
4. NO GLOVES While Spindle is Rotating

2. PRIMARY OPERATIONAL HAZARDS

1. Rotating spindle, collet, and cutting tool entanglement hazard
2. Flying hot chips and broken cutter ejection
3. Pinch hazard between moving table and knee saddle

3. BEFORE OPERATION CHECKLIST

1. Check drawbar engagement and verify collet/tooling is properly seated.
2. Set speed via variable speed dial ONLY while spindle is running, or shift Hi-Neutral-Lo range lever ONLY when spindle is completely stopped.

4. DURING OPERATION SAFETY RULES

1. Verify quill lock and table locks are set appropriately for the cut.
2. Confirm climb vs. conventional milling setup (use conventional milling on manual leadscrews to prevent cutter grab).
3. Never reach around a rotating cutter or adjust workpiece while spindle is turning.

5. AFTER USE & SHOWROOM CLEANUP

1. Return machine to showroom condition (brush chips with shop brush-NEVER use compressed air).
2. Home and wipe down table, ways, and vise.
3. Remove collets/tooling from spindle and return to designated tool rack.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Turn rotary drum switch to center OFF position or disengage main disconnect lever on right fence post; notify Lab Supervisor or SRA immediately.